

HW1

$$1. \quad \sigma = q(N_n + N_p) n_i$$

$$1000 = 1.6 \times 10^{-19} (100 + 50) n_i$$

$$n_i = 4.17 \times 10^{19}$$

$$n_i^2 = B T^3 e^{\frac{-E_g}{kT}}$$

$$(4.17 \times 10^{19})^2 = (1.08 \times 10^{31}) T^3 e^{-\frac{1.12}{8.62 \times 10^{-5} T}}$$

$$T \approx 2702 > 1430$$

intrinsic silicon will melt before reaching $\sigma = 1000$

$$2.(a) \quad p = N_A = 10^{16}$$

$$\sigma = q p N_p$$

$$= 1.6 \times 10^{-19} \times 1 \times 10^{16} \times 405$$

$$= 0.648$$

$$1(b) \quad n = p - N_A = N_D - N_A$$

$$\sigma = q \mu (N_D - N_A)$$

$$\sigma = q (N_D - N_A) \left[92 + \frac{1270}{1 + \left(\frac{N_A + N_D}{1.3 \times 10^{17}} \right)^{0.91}} \right]$$

$$5.5 = 1.6 \times 10^{-19} (N_D - 10^{16}) \left[92 + \frac{1270}{1 + \left(\frac{10^{16} + N_D}{1.3 \times 10^{17}} \right)^{0.91}} \right]$$

$$N_D = 4.58 \times 10^{16}$$

[illegible]